



UPSC Prelims

Science & Tech

Class 4

Applications of Biotechnology

Class 4 : topics to be covered



- GM crops and Transgenic Animals
- RDT in Medicine : Human Insulin, Gene Therapy
- Crispr CAS – 9
- Animal Cloning
- Stem Cell Therapy
- Vaccination
- Various Applications of Biotechnology

GM crops in India



- GEAC then MoEFCC
- DMH-11 : GM mustard : GEAC vacillating
- BT brinjal : immunity / diversity / side effects (33)
- Issue of Pest Resistance
- BT brinjal was banned
- New issues with Bt cotton : white fly attacks
- MS Swaminathan : called BT crops a failure, govt should resort to Genetic Engineering as a last resort only.

GM plants Regulation : Global



- Protocols to the Convention on Biological Diversity :
- Nagoya protocol
- Cartagena Protocol
- Nagoya Kuala Lumpur protocol
- We will study these when we study environment

GM Crops in India



- Regulatory Framework : Guided by the Environment Protection Act, 1986 and the Rules for the Manufacture/Use/Import/Export and Storage of Hazardous Microorganisms, Genetically Engineered Organisms or Cells, 1989 under the act
- Rules are broad in scope and cover most elements relating to GM crops including sale, storage, export, import production etc.
- Many committees are there for various functions relating to GM crops.

Competent Authorities



1. The Recombinant DNA Advisory Committee (RDAC)

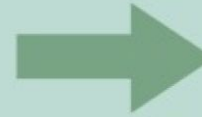


Advisory

2. Institutional Biosafety Committee (IBSC)

3. Review Committee on Genetic Manipulation (RCGM)

4. Genetic Engineering Appraisal Committee (GEAC)



Approval

5. State Biotechnology Coordination Committee (SBCC)

6. District Level Committee (DLC)



Monitoring

Various Committees



- Recombinant DNA Advisory Committee : under DBT, review developments in biotechnology at national and international levels.
- Institutional BioSafety Committee : at every institute, ensures that research in GM crops happens safely. Approves low risk experiments.
- Review Committee on Genetic Modification : Regulatory body inside DBT, responsible for bringing out manuals/ guidelines specifying procedures for research/ use /applications in GMOs. Reviews all projects involving high risk field experiments.
- Genetic Engineering Appraisal Committee : Apex regulatory committee under MOEFCC and gives approval for commercial release of GM crops.

Various Committees



- State Biotechnology Coordination Committee: Monitors at the state level that activities relating to GM crops happen as per regulations. State level nodal agency to assess damage due to GM crops.
- District Level Committee : Monitors at the district level activities related to GM crops. Headed by DC.

Various Committees



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GEAC : Most important body



- Import, export, manufacture, store, process, use, sell GMOs only with the approval of GEAC.
- Under MoEFCC
- It can also revoke approvals in case of :
 - New information on harmful effects of GMOs
 - GMOs causing damage to environment as was not seen at time of approval
 - Non Compliance of any conditions stipulated by GEAC.

Genetic Engineering Appraisal Committee (GEAC)

Functions under: Ministry of Environment, Forest and Climate Change (MoEF&CC).

Objectives: Responsible for approval of

- ⇨ **Large scale use of hazardous microorganisms and recombinants** in research and industrial production from environmental angle.
- ⇨ **Release of genetically engineered organisms and products into environment** including experimental field trials.
- ⇨ Taking **punitive action under the Environment (Protection) Act 1986**.

Composition: Presently, it has **24 members** and it meets every month. **Chaired by Special Secretary/Additional Secretary of MoEF&CC.**



Exemption from the rules



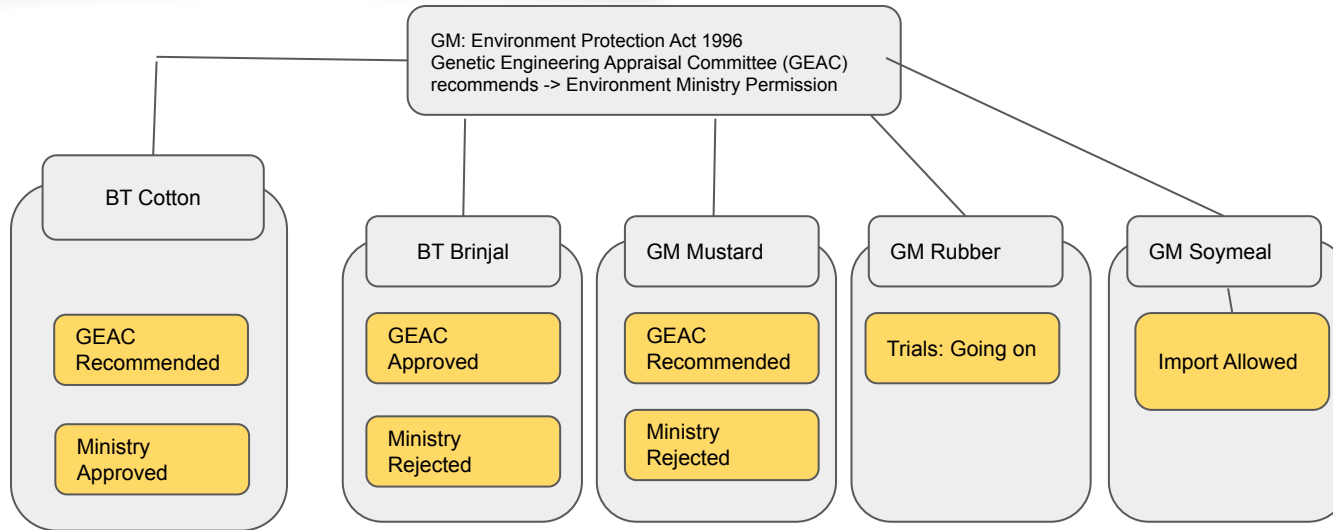
- MoEFCC can grant exemption to certain GMO handlers from the provisions of the rules
- Exemption can be granted for :
 - Indigenous development of recombinant pharmaceuticals that have less risk (risk group 1 and 2)
 - Import and marketing of products derived from LMOs where the end product is not an LMO.
 - Event based approval mechanism for Bt Cotton Hybrids : BT cotton hybrids expressing approved events developed through conventional hybridization are exempted from conduct of detailed biosafety trials.
- Recently, GEAC has given approvals for biosafety trials of Event 142 : a new variety of Bt Brinjal.

GM crop Regulation in India

Mandate of Ministries/Departments

Ministry of Environment, Forest and Climate Change	- Primarily responsible for conservation and protection of environment, ensuring environmental and human health safety before release of GMOs / LMOs. - Nodal agency for implementing Rules, 1989 and the Cartagena Protocol on Biosafety
Department of Biotechnology (Ministry of Science & Technology)	- Nodal department for promoting biotechnology programs - Provides scientific support in implementation of biosafety regulations - Provide services in areas of research, infrastructure, generation of human resource
Ministry of Agriculture	- Policies aimed at agriculture growth. - Indian Council of Agricultural Research (ICAR) responsible for monitoring agronomic benefits of GM technology. - Monitoring post-release performance of GM crops.
Ministry of Health and Family Welfare	- Policies aimed at protecting and monitoring human health. - Food Safety and Standards Authority of India responsible for regulating genetically engineered foods.
Ministry of Commerce and Industries	- Enhance trade with other countries through export/import policies. - Nodal agency for implementing DGFT notification on GMOs
Central Board of Excise and Customs, Department of Revenue, Ministry of Finance	Enforcement of regulation pertaining to transboundary movement of GMOs/LMOs at point of entry

GM Crops in India



GM crops in India



- Import of GM soymeal allowed : GEAC was bypassed as MoEFCC gave exemption on the basis of the final product being non living. Soyameal is protein rich solid leftover raw material after extracting oil from soybean seed. Major ingredient of poultry feed.
- GM rubber : trials going on. World's first GM rubber plant was planted in assam. Developed specifically for the northeast. Additional copies on gene MnSOD : Manganese Containing superoxide dismutase. This helps plant tide over cold conditions of winter. Developed by kerala based rubber research institute of india.
- Natural rubber : grows in amazon rainforest where there is no cold season and climate is hot and humid.

GM crops in India



- Agharkar research institute (under DST) has mapped 2 genes in wheat called Rht14 and Rht18 : these lead to better seedling vigour and longer coleoptiles (protecting sheath)
- This can allow deeper planting of wheat seeds and can reduce crop residue burning.
- GM food imports : GEAC clears from environmental angle and FSSAI clears from food consumption human health angle. So GM imports are regulated under both EPA, 1986 and Food safety and Standards act, 2006.

2023 current affairs



- DBT has issued guidelines for safety assessment of Genome edited Plants, 2022 easing norms for research into GM crops.
- They have exempted Genome edited plants falling under category of SDN-1 and SDN-2 from 1989 rules. This exemption was earlier granted by MoEFCC.
- These guidelines apply to plants that are genome edited using SDN.
- SDN = Site directed nuclease. It is a type of Gene editing technology. We will study it in detail soon. Word breakdown : Similar to Restriction Endonuclease.

GM plants



- Flavr Savr Tomato : first commercial GM food crop
- Golden Rice : Provitamin A
- BT Crops : Cry 1 AC gene : Protection against Bollworms



Genome editing, Transgenesis and Genetic Modification



- Genome editing : Making changes in the genome but no introduction of foreign genetic material. Only 'cuts' are made, using enzymes such as Site Directed Nuclease and Site Specific Nuclease.
- Genetic Modification : Modification of Organism using Genetic Material from a foreign source. External (alien) DNA is added.
- Transgenesis and Genetic Modification are mostly similar. There is a small difference.
- Although both have altered genomes, a transgenic organism is a GMO containing a DNA sequence or a gene from a different species. Thus, all transgenic organisms are GMOs, but not all GMOs are transgenic.

Recombinant DNA Tech



- Introduction of foreign DNA into host
- Recombinant Human Insulin : Eli Lilly
- Recombinant HGH
- Recombinant Vaccines : we will study in detail
- Recombinant Factor VIII : Haemophilia
- Golden Rice : not currently in use
- Crops

Transgenic Animals



Animals that have had their DNA manipulated to possess and express an extra (foreign) gene are known as transgenic animals.

Why?

<i>normal physiology and development</i>	<i>Study of disease</i>	<i>Biological products : example : Rosie and lactalbumin milk</i>	<i>Vaccine safety</i>	<i>Chemical safety testing</i>
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Transgenic Animals : Applications



- Human alpha 1 anti tripsin : in sheep
- Transgenic pigs : tests for organ transplantation

RDT applications in Medicine

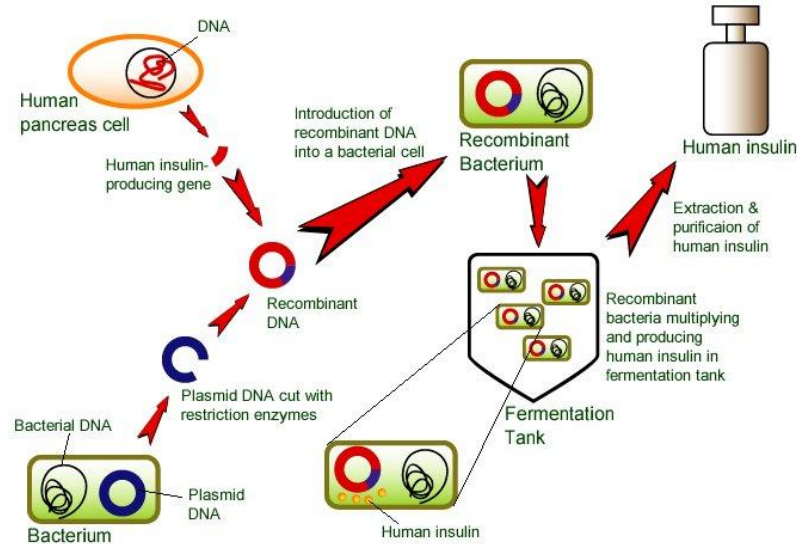


- Management of Adult onset diabetes : earlier we had to take insulin from other animals by killing them.
- This lead to unwanted immune responses (more in the class on immunity)
- So we needed human insulin. But we couldn't kill other humans right? :P
- Solution : Genetic engineering of bacteria to induce them to produce human insulin.
- Produced by Company called Eli-Lily in 1983.

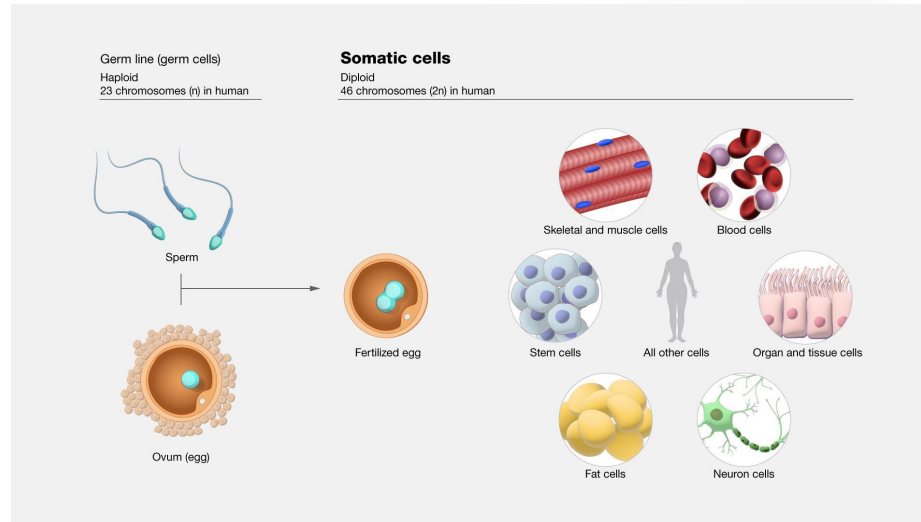
Genetically Engineered Insulin



Human Insulin Production



BASICS : Somatic and Germline cells



Gene Therapy



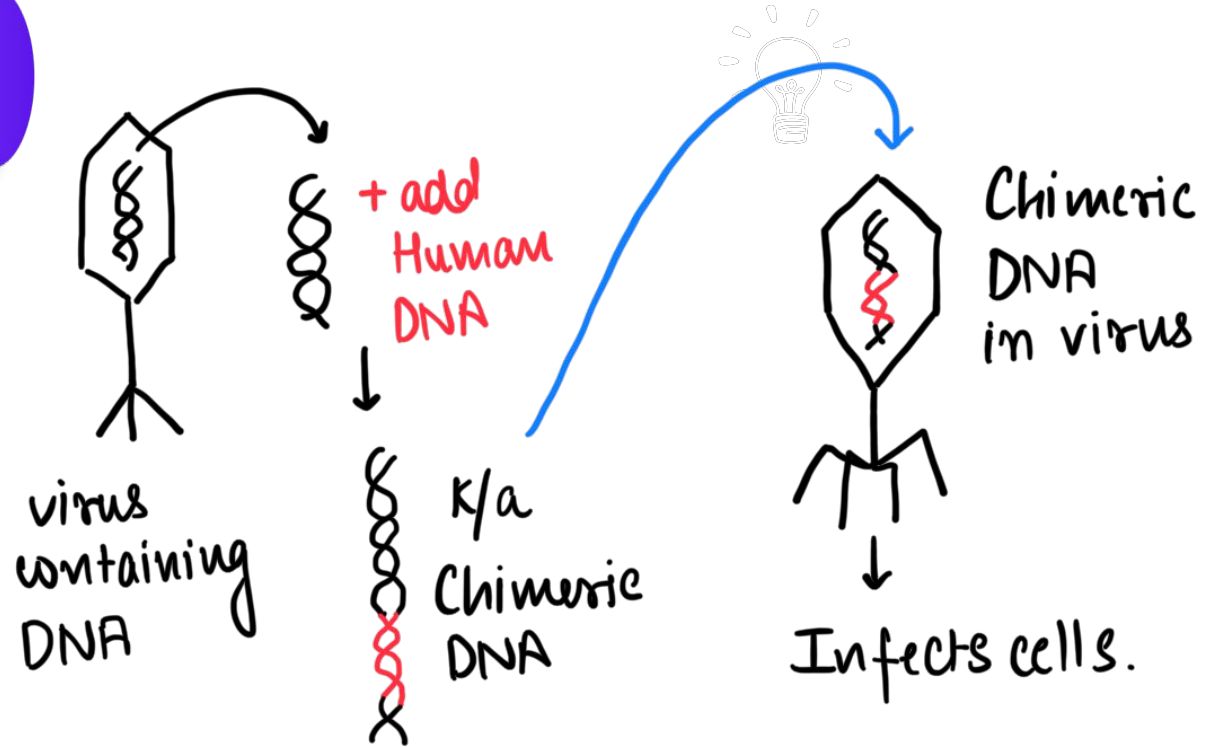
- A child is born with genetic defects. One important gene that he has is non functional. What do we do?
- Delivery of normal genes through baby with gene therapy.
- Gene therapy is a collection of methods that allows correction of a gene defect that has been diagnosed in a child/embryo.
- Here genes are inserted into a person's cells and tissues to treat a disease.
- Correction of a genetic defect involves delivery of a normal gene into the individual or embryo to take over the function of and compensate for the non-functional gene.

Gene Therapy

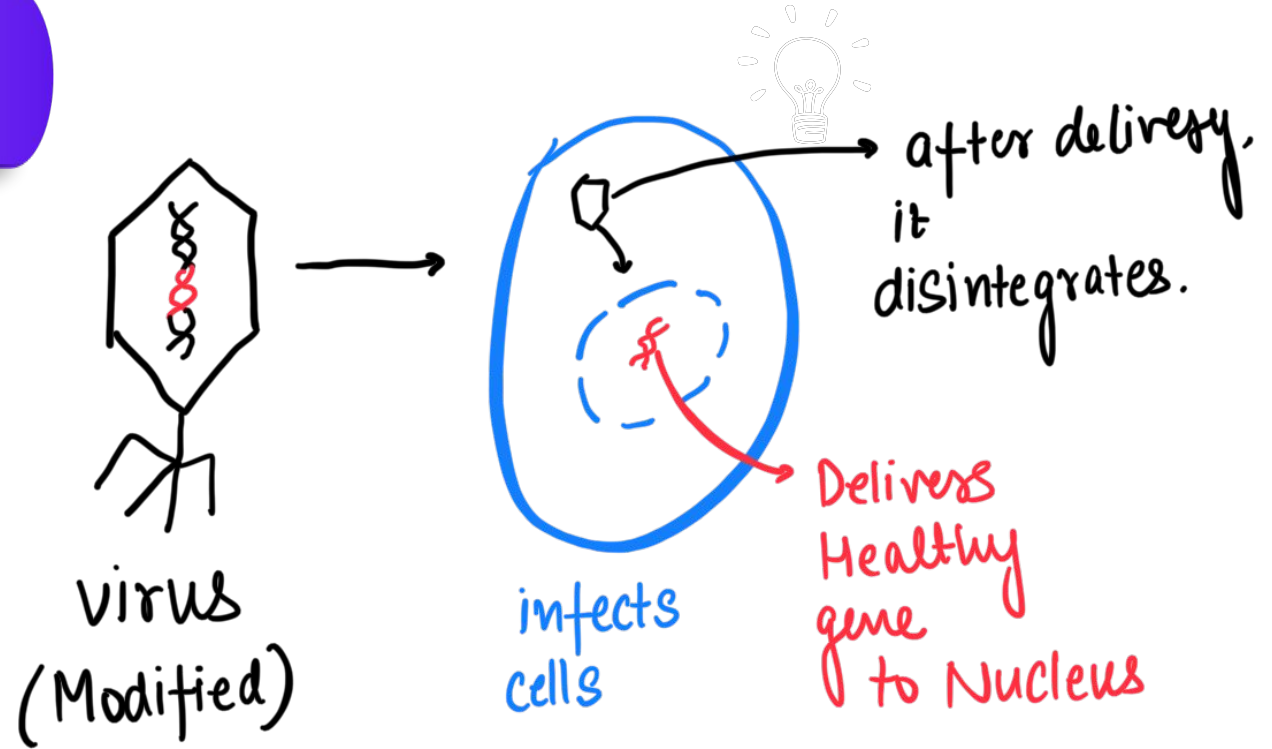


- Biotechnological method to replace defective gene with correct functioning gene.
- 2 types : Somatic cell gene therapy and Germline Gene therapy
- Somatic Gene therapy : Genetic Disorders in individuals can be cured.
- Ex : Haemophilia, Sickle Cell Anaemia (important this year : WHY?)
- Germ Cell Gene therapy : Offspring (child) can be prevented from getting disease.
- Can be used to fight cancer as well : CAR-T cell therapy.

How is it done ?



How is it done (2)



LET'S SEE A VIDEO ON GENE THERAPY



- [click here](#)
- [click here for video 2](#)

Gene Therapy



Can be used to :

- 1. Replace bad gene with healthy gene
- 2. Add a new gene to perform function of missing / non functional gene
- 3. Turn off a problem gene.
- Can be done inside the body (using virus) called in vivo
- We are using modified human virus, for example adenovirus.
- Can be done outside the body : in vitro

In vitro gene therapy : Example : ADA - SCID

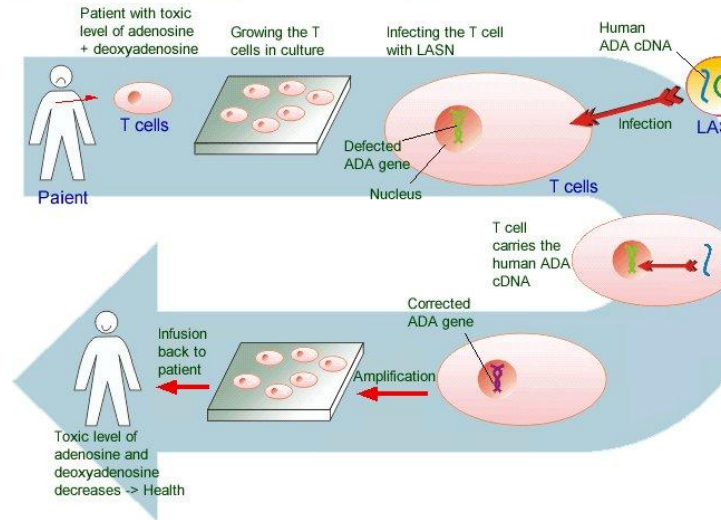


- Adenosine Deaminase Deficiency leading to severe combined immunodeficiency
- T cells of the body need ADA.
- When ADA is absent, patient develops SCID.
- We need to deliver ADA to patient's T cells.
- We use functional ADA gene inserted with viral vector. But virus has to be genetically edited to be made safe.
- T cells are first removed from the patient, edited with viral vector containing new gene, then sent back into the patient.

Gene therapy for ADA- SCID



Gene Therapy for ADA-SCID

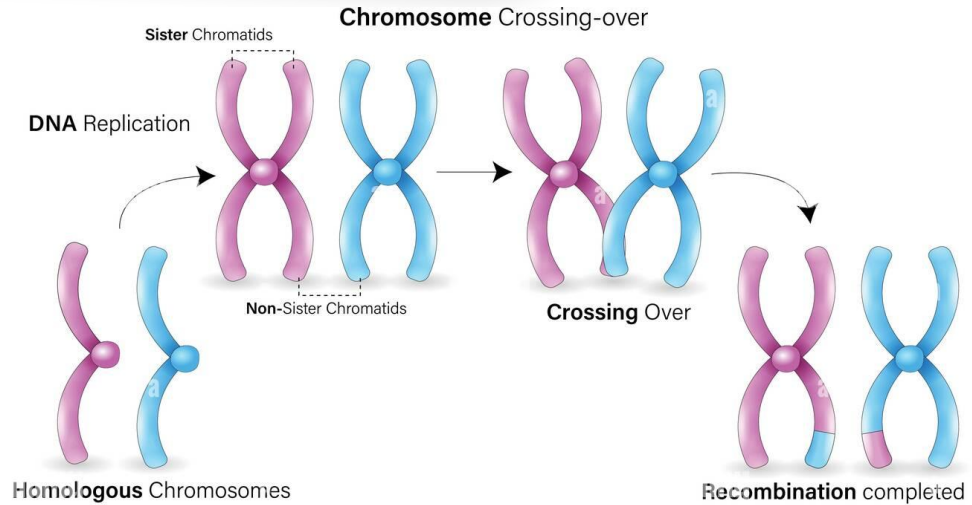
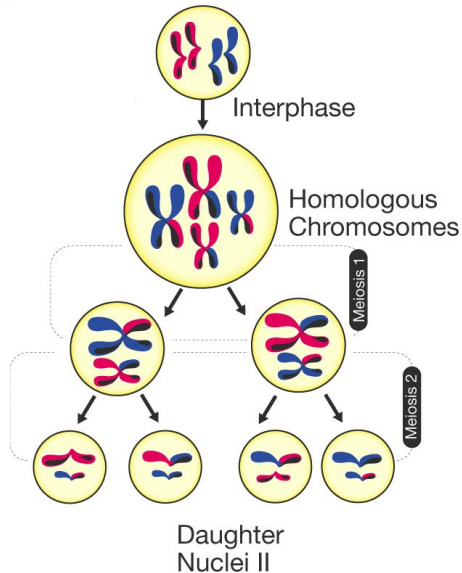


Issues of Gene Therapy



- Short Lived Nature : multiple rounds needed : example : ADA SCID
- Possible solution : Do at early stage.
- Immune response possible, other side effects possible.
- Cannot be used to correct chromosomal disorders, only disorders with single bad gene. (explained later)
- Virus alteration and manipulation : Containment

Chromosomal Disorders : defects during mieosis

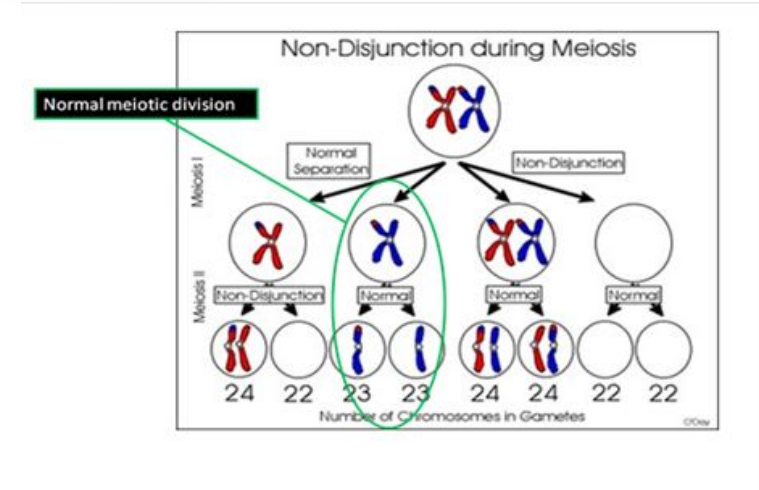


Chromosomal Recombination

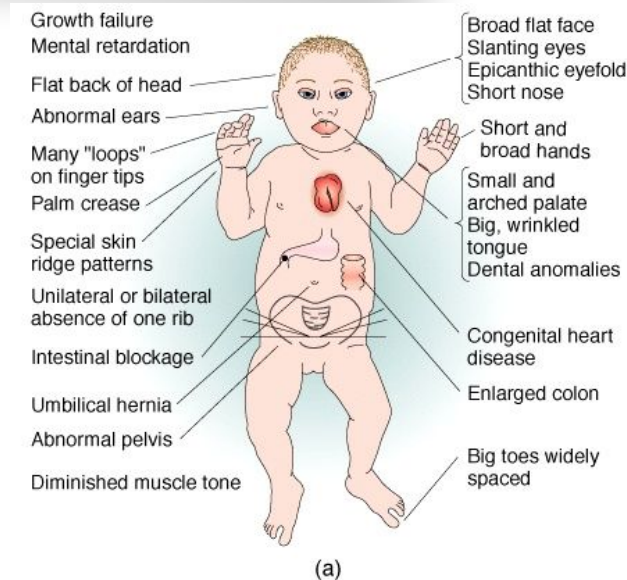
Chromosomal disorders : Down's Syndrome



- Patient has 3 copies of Chromosome number 21
- Leads to severe genetic defects.
- This cannot be cured by gene therapy.



Down's Syndrome : treat them with sympathy

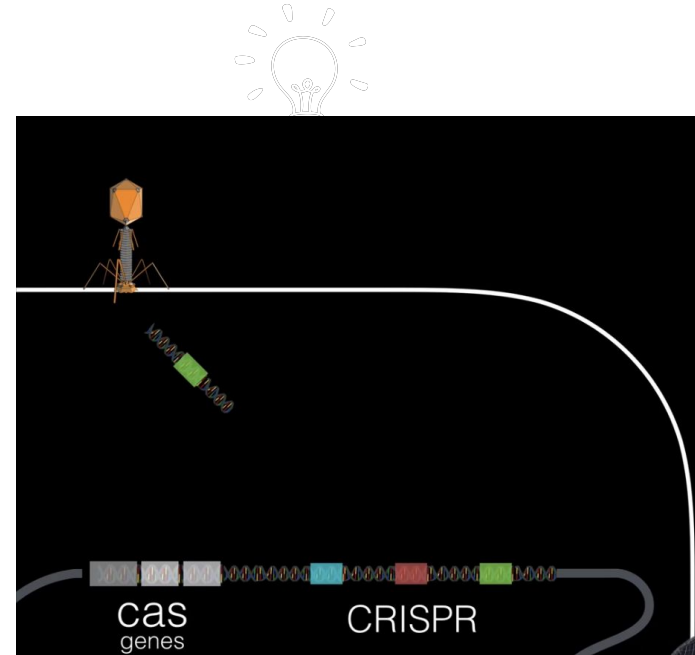


Crispr Cas – 9 and Gene Editing

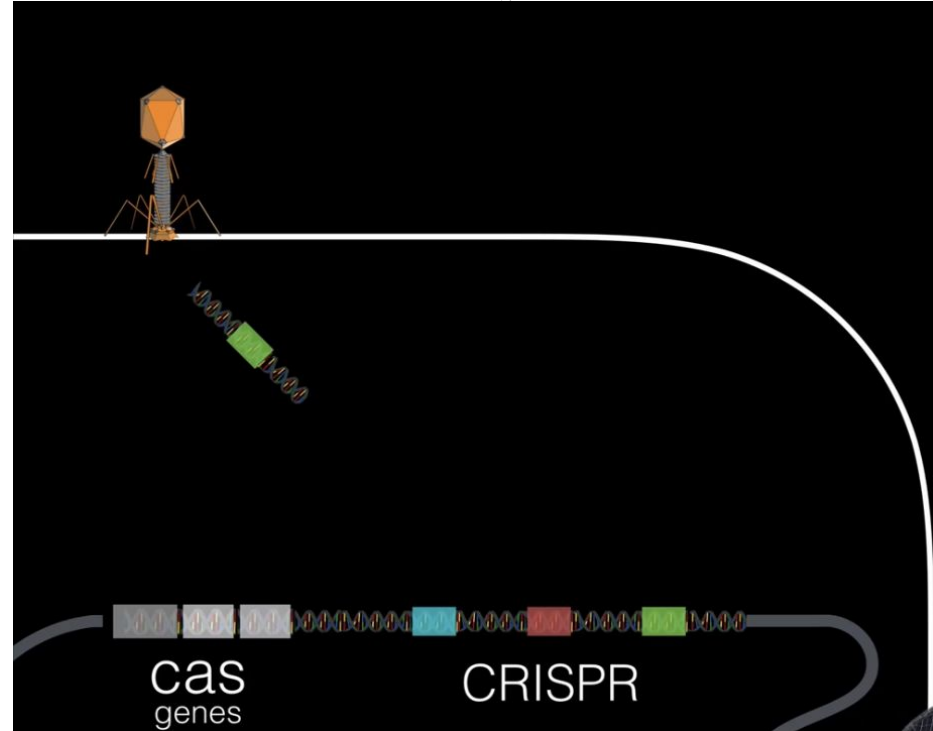


- What are genes?
- Gene Editing : changing genes in humans
- CRISPR : Clustered Regularly Interspaced short Palindromic Repeats
- CAS : crispr associated protein -9
- Originally found in bacteria
- Molecular level scissors and glue (cut and paste mechanism)

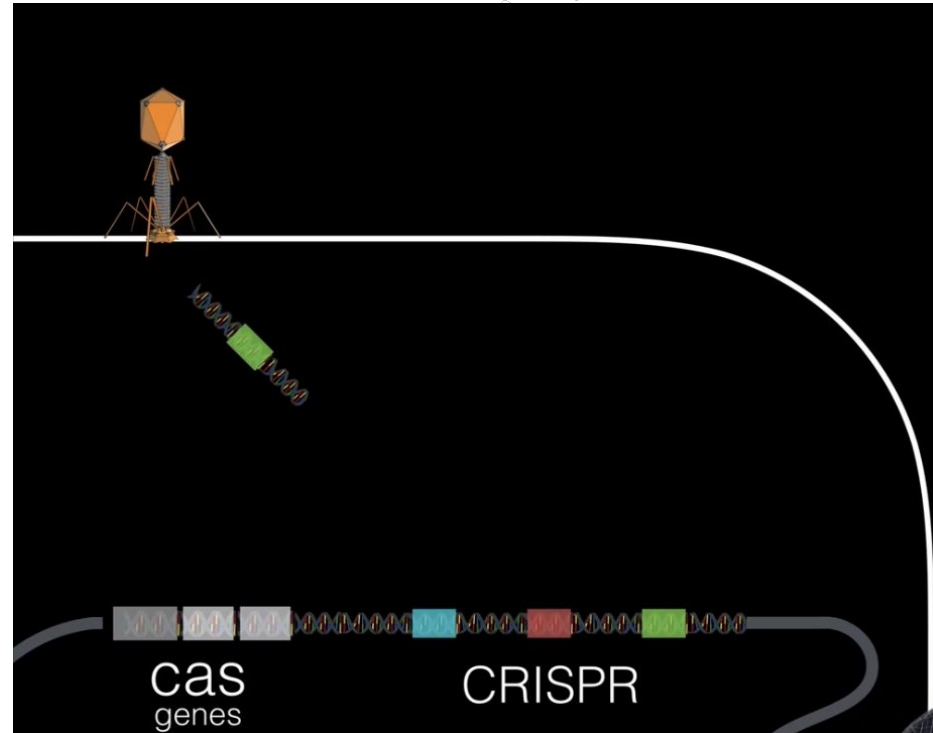
- Palindromes interspaced by spacer DNA
- Cas gene : Cas proteins : Helicases (unwind DNA) Nucleases (Cut DNA)
- It is a defence system that bacteria uses to defend against viruses that infect bacteria. These viruses are called bacteriophage.



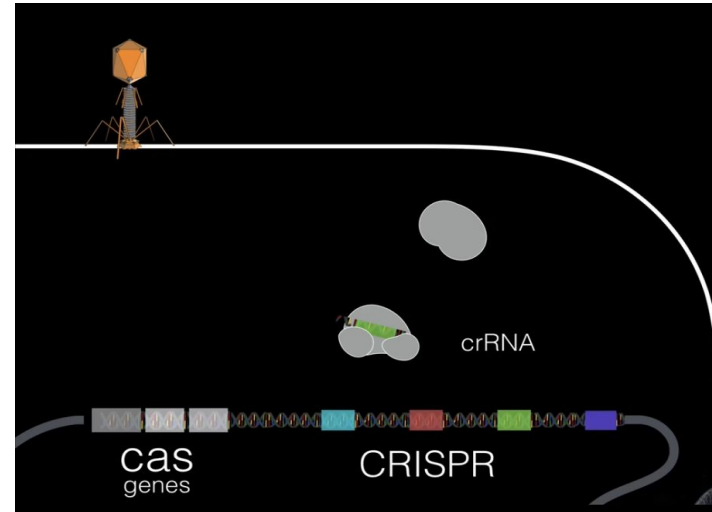
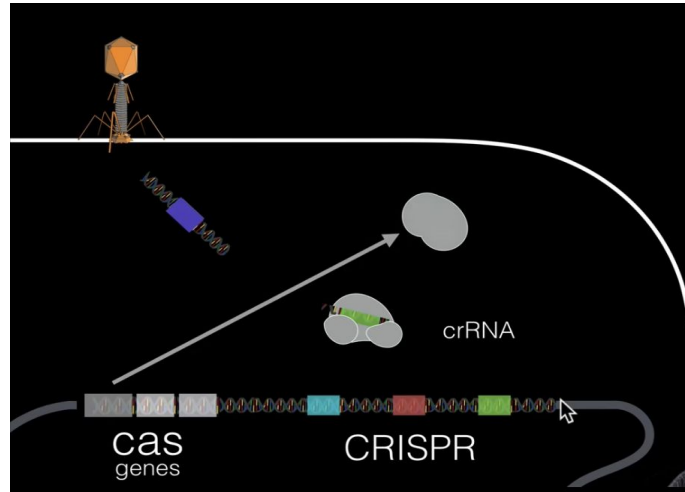
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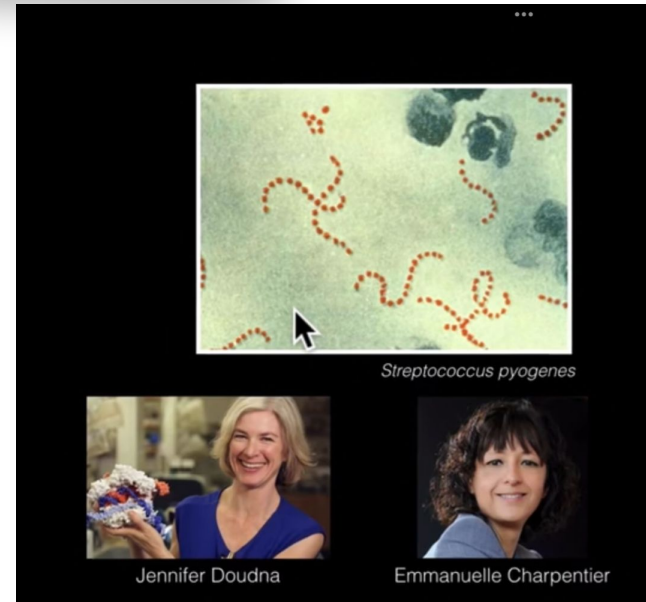
Crispr Cas-9 : Working explained



Crispr



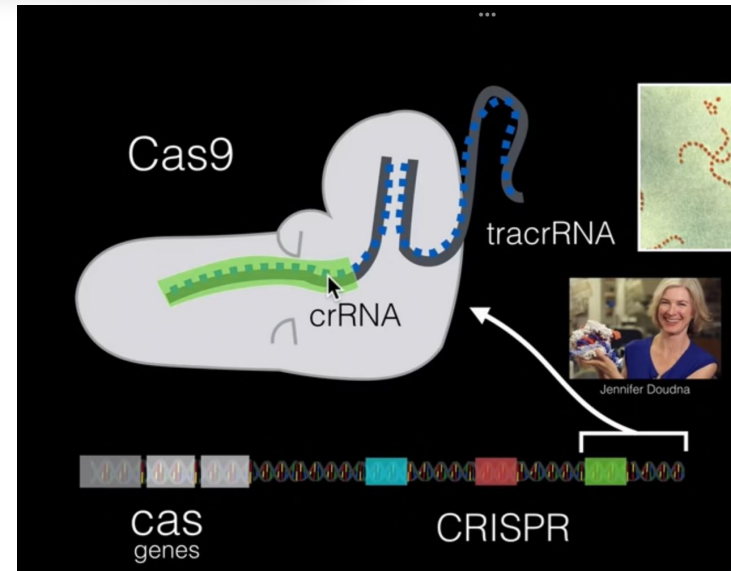
- Spacer – repeat – Spacer– Repeat
- Similar to our immune system.
- Scientists thought of a new use for this in genetic editing.



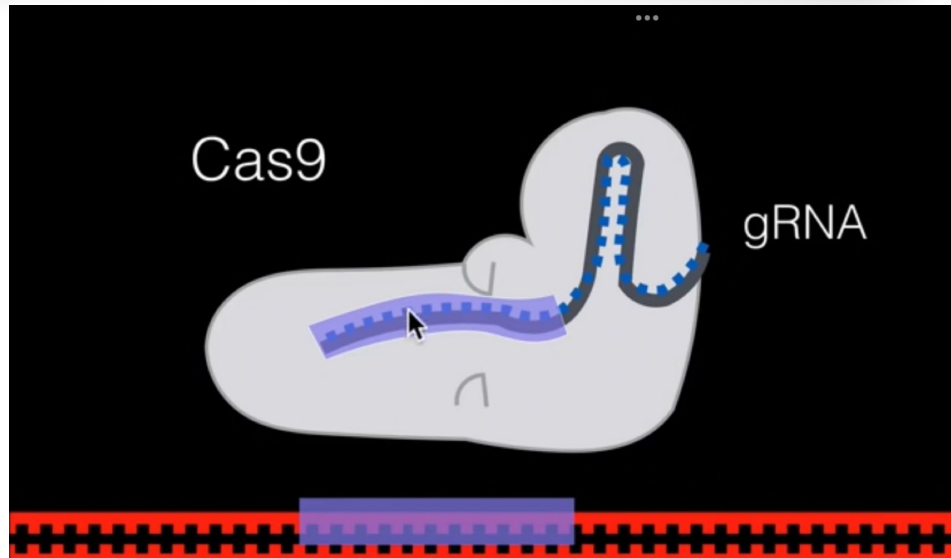
Crispr in genetic editing



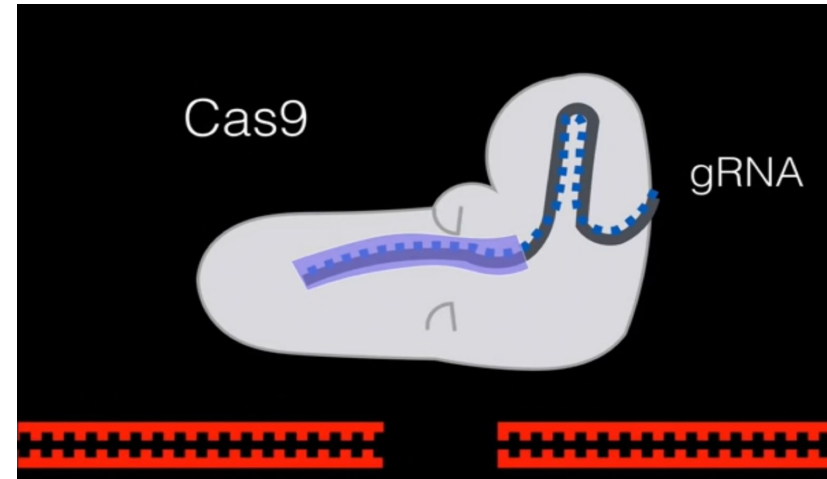
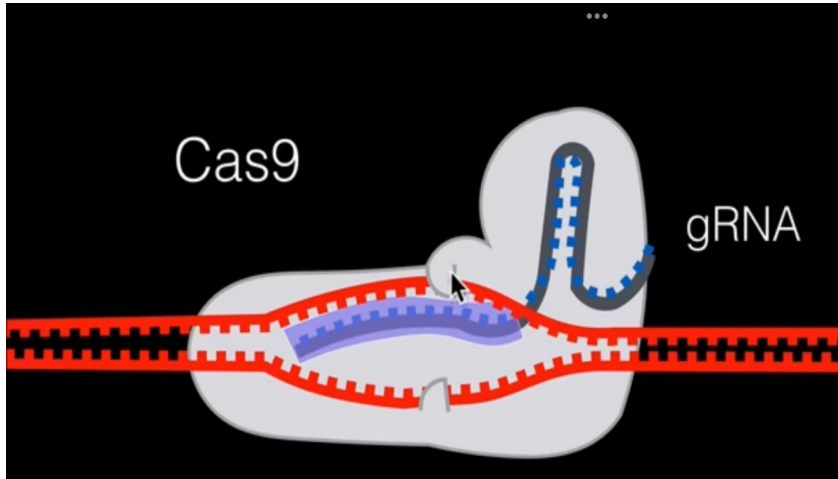
- Nuclease : enzyme responsible for cutting DNA



We modified CRISPR to work in our cells.



We modified CRISPR to work in our cells.



CRISPR Cas 9



- Consists of single guide RNA (sgRNA) : complementary to the part of DNA to be cut
- Cas 9 : crispr associated protein 9 : cuts specific part of dna
- It is like using Microsoft word for a document
- Ctrl + F = Find (SgRNA)
- Replace = Delete and change (Cas-9)
- Crispr cas 9 is molecular scissors

Possible Applications



- Gene editing and gene therapy
- Creation of Genetically Modified Organisms
- Designer babies : Chinese Scientist Jailed for this
- Germline editing of reproductive cells

Issues



- Increased risk of mutations and cancer
- Ethical concerns and designer embryos
- Impact on human variation
- Abuse of gene editing
- Safety and off target effects
- Informed Consent
- Justice and equity

Gene Doping



Changing DNA to enhance performance by athletes

Ex vivo (in vitro)

In vivo

WADA issues

Animal Cloning



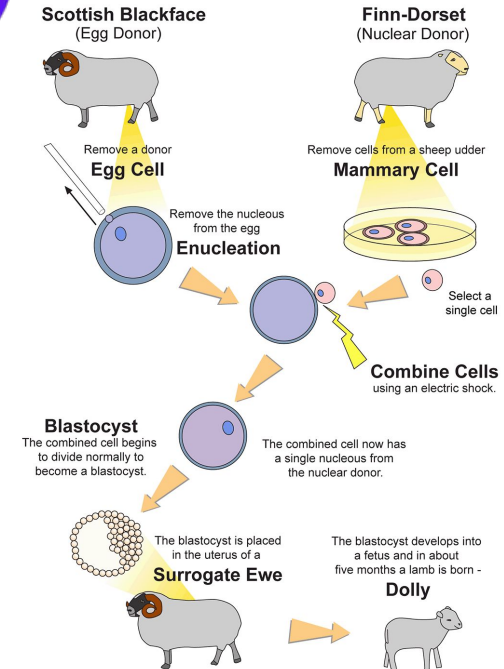
- Identical twins : from 2 parents
- Cloning is the development of cloned animal from a single parent. Also known as Reproductive Cloning.
- Method : Somatic Cell Nuclear Transfer Technique

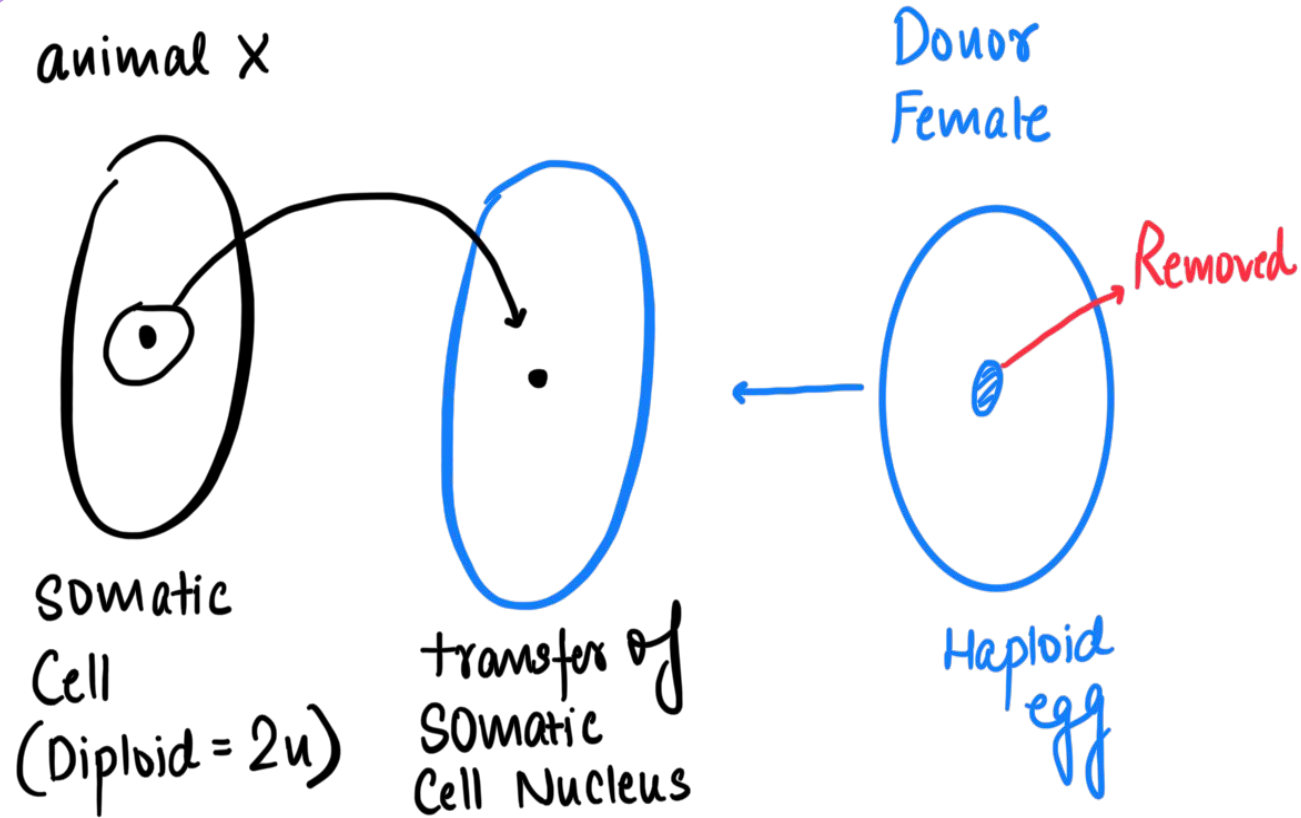
Animal Cloning and Somatic Cell Transfer



- Cloning : exact same genes of progeny
- Dolly the sheep : Ian Wilmut
- Reproductive cloning
- We will study **Therapeutic Cloning after stem cells**

The Story of Dolly





Immunity



- We get exposed to many disease causing pathogens daily, but this does not cause disease everytime
- Body has a system to defend against foreign organisms
- Remember : Was there such a system in bacteria?

Try doing this question



Q. Consider the following statements: DNA Barcoding can be a tool to:

1. assess the age of a plant or animal.
2. distinguish among species that look alike.
3. Identify undesirable animal or plant materials in processed foods.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 3 only
- (c) 1 and 2
- (d) 2 and 3

Types of Immunity : Innate and Acquired



- Innate Immunity
- non-specific type of defence, that is present at the time of birth
- Innate immunity consist of four types of barriers.
- Physical barriers : skin / mucus
- Physiological barriers : acid in stomach
- Cellular barriers : WBC (white blood cells) : B cells and T cells
- Cytokine barriers : Interferon

Acquired Immunity

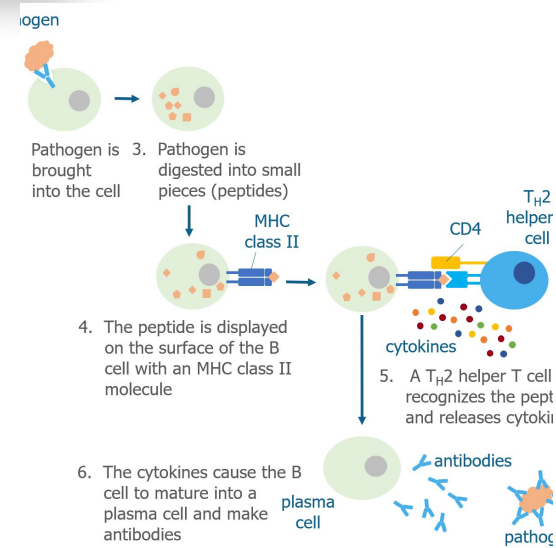


- Why were you able to solve the DNA barcoding question?
- You had done it before : effect of memory : same principle works in immune system
- Acquired immunity : pathogen specific, works on the principle of memory
- our body when it encounters a pathogen for the first time produces a response called primary response which is of low intensity. Subsequent encounter with the same pathogen elicits a highly intensified secondary or anamnestic response. This is ascribed to the fact that our body appears to have memory of the first encounter.
- primary and secondary immune responses are carried out with the help of two special types of lymphocytes present in our blood, i.e., B-lymphocytes and T-lymphocytes

B cells



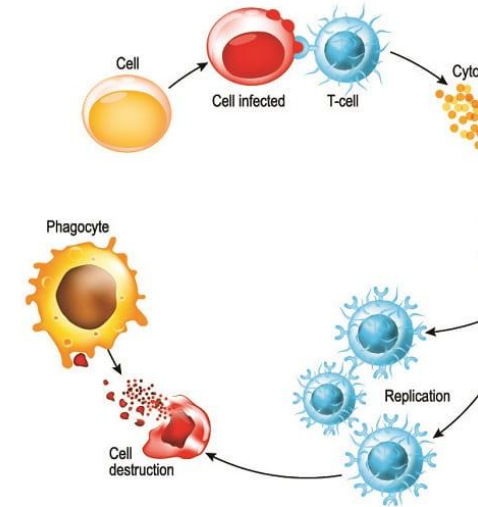
- The B-lymphocytes
- produce an army of proteins in
- response to pathogens into our
- blood to fight with them. These
- proteins are called antibodies.



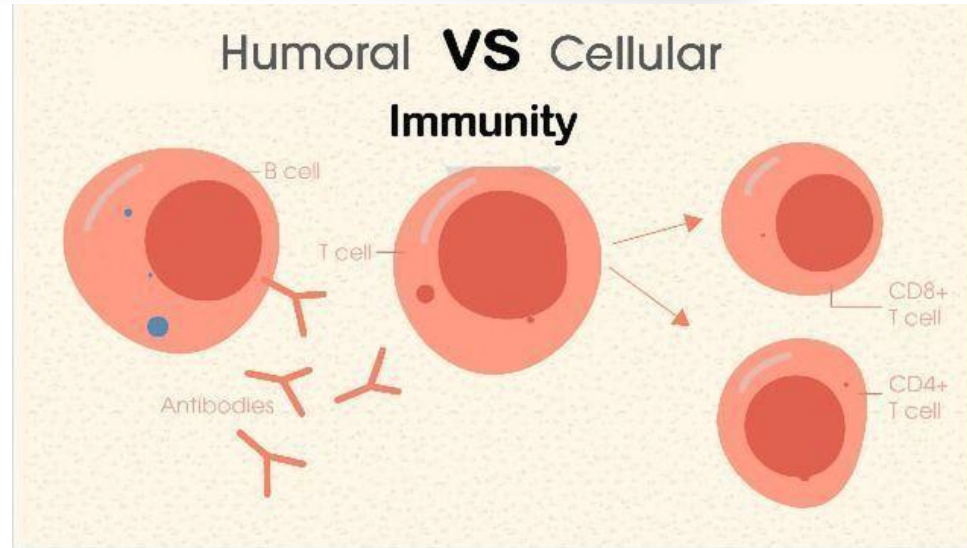
T cells



- T-cells themselves do not secrete antibodies but help B cells produce them. Each antibody molecule has four peptide chains, two small called light chains and two longer called heavy chains. Hence, an antibody is represented as H2L2.
- Because these antibodies are found in the blood, the response is also called as humoral immune response.
- The second type is called cell-mediated immune response or cell-mediated immunity (CMI). The T-lymphocytes mediate CMI.



Humoral and Cell Mediated Immunity

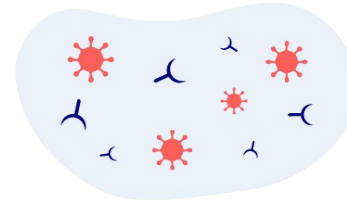


Active vs Passive Immunity



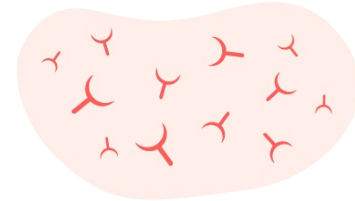
- When a host is exposed to antigens, which may be in the form of living or dead microbes or other proteins, antibodies are produced in the host body. This type of immunity is called active immunity.
- Active immunity is slow and takes time to give its full effective response.
- When ready-made antibodies are directly given to protect the body against foreign agents, it is called passive immunity.
- Ex : Mother's milk and colostrum (IgA)

ACTIVE IMMUNITY



Administering living (weakened) or dead pathogens creates a protective immune response

PASSIVE IMMUNITY



The transfer of antibodies from an immunized donor to a non-immune recipient

Auto Immunity

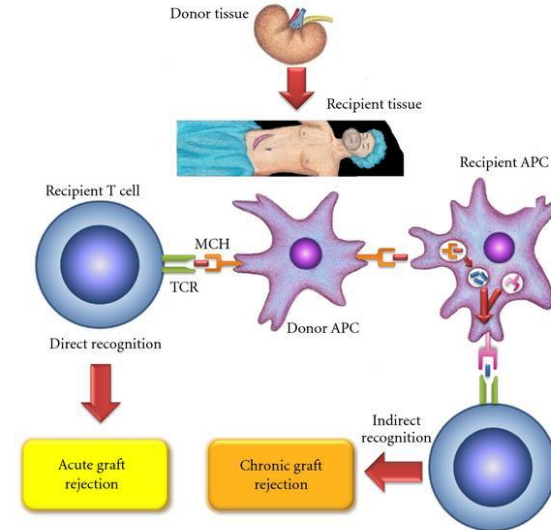


- Memory-based acquired immunity evolved in higher vertebrates based on the ability to differentiate foreign organisms (e.g., pathogens) from self- cells
- sometimes, due to genetic and other unknown reasons, the body attacks self-cells.
- This results in damage to the body and is called auto-immune disease. Ex :Rheumatoid arthritis

Immune system and Grafts



- Graft rejection happens due to our own immune system.
- What could be the possible solution?
- We have read some technologies that help prevent this, try and recall.



Vaccination



- Why am I making you solve prelims questions in a controlled environment with guidance?
- If you can solve easy questions in this setting, you will be able to solve tough ones also in a difficult setting later as you would have developed a certain memory and technique
- Same exact principle applies in the case of Vaccination : based on the property of 'memory' of the immune system.

Active Vaccination



- In vaccination, a preparation of antigenic proteins of pathogen or inactivated/weakened pathogen (vaccine) are introduced into the body.
- The antibodies produced in the body against these antigens would neutralise the pathogenic agents during actual infection.
- The vaccines also generate memory – B and T -cells that recognise the pathogen quickly on subsequent exposure and overwhelm the invaders with a massive production of antibodies.

Passive Vaccination



- If a person is infected with some deadly microbes to which quick immune response is required as in tetanus
- we need to directly inject the preformed antibodies, or antitoxin (a preparation containing antibodies to the toxin).
- Even in cases of snakebites, the injection which is given to the patients, contain preformed antibodies against the snake venom. This type of immunisation is called passive immunisation.

Active vs Passive Vaccination



	Active vaccination	Passive vaccination
Immunization with	Antigen	Pre-formed antibodies
Required time until protection occurs	Weeks-months	Minutes-hours
Duration of protection	Ideally for a lifetime	Few weeks
Role of host immune system	Develops own protective immunity	Only effector mechanisms are activated by administered antibodies
Cells participating	Ideally all cells of the immune system	No cells are affected or only effector cells are activated
Induction of immunological memory	Yes	No

AEFI



- There is no such thing as a "perfect" vaccine which protects everyone who receives it AND is entirely safe for everyone.
- Effective vaccines (i.e. vaccines inducing protective immunity) may produce some undesirable side effects which are mostly mild and clear up quickly.
- EFIs can be related to the vaccine itself (product or quality defect-related reactions), to the vaccination process (error or stress related reactions) or can occur independently from vaccination (coincidental) and are classified as:
 - Vaccine product-related reaction
 - Vaccine quality defect-related reaction
 - Immunization error-related reaction
 - Immunization anxiety-related reaction
 - Coincidental event

Types of Vaccines



- Based on method of Administration
- Im
- Iv
- Sc
- Nasal
- Oral

Types of vaccines : based on pathogen



- Live Attenuated
- Inactivated
- Subunit
- Toxoid
- Recombinant
- mRNA
- DNA

Inactivated Vaccine



- Inactivated vaccines use the killed version of the germ that causes a disease.
- Inactivated vaccines usually don't provide immunity (protection) that's as strong as live vaccines. So you may need several doses over time (booster shots) in order to get ongoing immunity against diseases.
- Inactivated vaccines are used to protect against:
 - Hepatitis A
 - Flu (shot only)
 - Polio (shot only)
 - Rabies

Live Attenuated Vaccine



- Live vaccines use a weakened (or attenuated) form of the germ that causes a disease.
- Because these vaccines are so similar to the natural infection that they help prevent, they create a strong and long-lasting immune response. Just 1 or 2 doses of most live vaccines can give you a lifetime of protection against a germ and the disease it causes.
- But live vaccines also have some limitations. For example:
- Because they contain a small amount of the weakened live virus, they may cause disease in some people, especially those with a compromised immune system
- They need to be kept cool, so they don't travel well.
- Live vaccines are used to protect against:
- [Measles, mumps, rubella](#) (MMR combined vaccine)
- [Rotavirus](#)
- [Smallpox](#)
- [Chickenpox](#)
- [Yellow fever](#)

Subunit Vaccine



- A subunit vaccine is a vaccine that contains purified parts of the pathogen that are antigenic, or necessary to elicit a protective immune response.
- A "subunit" vaccine doesn't contain the whole pathogen, unlike live attenuated or inactivated vaccine, but contains only the antigenic parts such as proteins, polysaccharides or peptides.
- Because the vaccine doesn't contain "live" components of the pathogen, there is no risk of introducing the disease, and is safer and more stable than vaccine containing whole pathogens. Other advantages include being well-established technology and being suitable for immunocompromised individuals
- Disadvantages include being relatively complex to manufacture compared to some vaccines, possibly requiring adjuvants and booster shots, and requiring time to examine which antigenic combinations may work best.

Recombinant Viral Vector Vaccine



- Uses a safe virus to deliver specific sub parts of a pathogen.
- Ex : Spike protein of coronavirus
- Immune response is generated without causing a disease
- Genetic instructions of making such a protein are inserted into a safe virus (through RDnaTech)
- Several different viruses have been used as vectors, including influenza, vesicular stomatitis virus (VSV), measles virus, and adenovirus, which causes the common cold. Adenovirus is one of the viral vectors used in some COVID-19 vaccines being studied in clinical trials.

Conjugate vaccine



- A conjugate vaccine is a type of subunit vaccine which combines a weak antigen with a strong antigen as a carrier so that the immune system has a stronger response to the weak antigen.
- Most vaccines contain a single antigen that the body will recognize. However, the antigen of some pathogens does not elicit a strong response from the immune system, so a vaccination against this weak antigen would not protect the person later in life.
- In this case, a conjugate vaccine is used in order to invoke an immune system response against the weak antigen. In a conjugate vaccine, the weak antigen is covalently attached to a strong antigen, thereby eliciting a stronger immunological response to the weak antigen.
- Most commonly, the weak antigen is a polysaccharide that is attached to strong protein antigen. However, peptide/protein and protein/protein conjugates have also been developed.

Polysaccharide Vaccine



- Polysaccharides : poly carbohydrates
- Vaccines that consist of Poly carbohydrate molecule that generates immune response
- Generally vaccines are made of protein, that's why polysaccharide vaccines are special.

mRNA vaccine



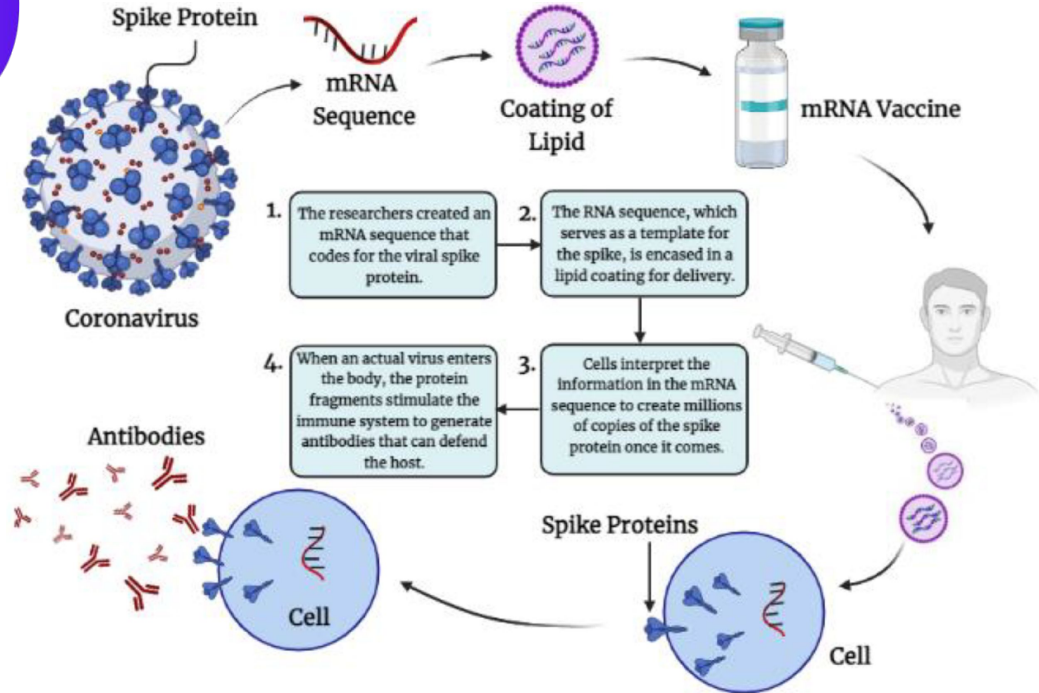
- What is mRNA?
- What is the vaccine principle?
- Now you try and tell me what a mRNA vaccine will do.

mRNA Vaccines : type of nucleic acid vaccine



- An mRNA vaccine is a type of vaccine that uses a copy of a molecule called messenger RNA (mRNA) to produce an immune response. The vaccine delivers molecules of antigen-encoding mRNA into immune cells, which use the designed mRNA as a blueprint to build foreign protein that would normally be produced by a pathogen (such as a virus) or by a cancer cell. These protein molecules stimulate an adaptive immune response that teaches the body to identify and destroy the corresponding pathogen or cancer cells.

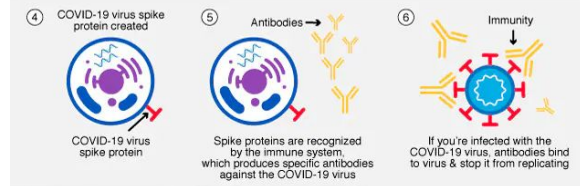
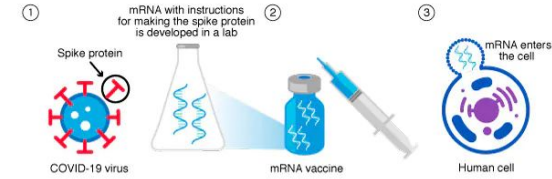
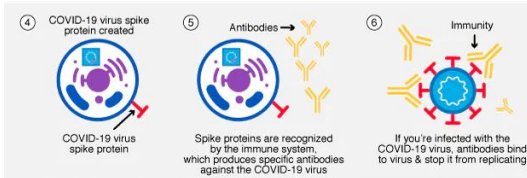
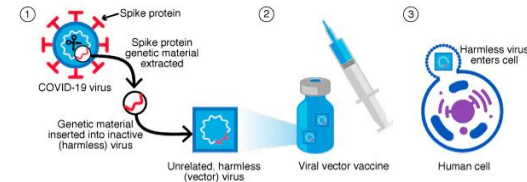
mRNA vaccine

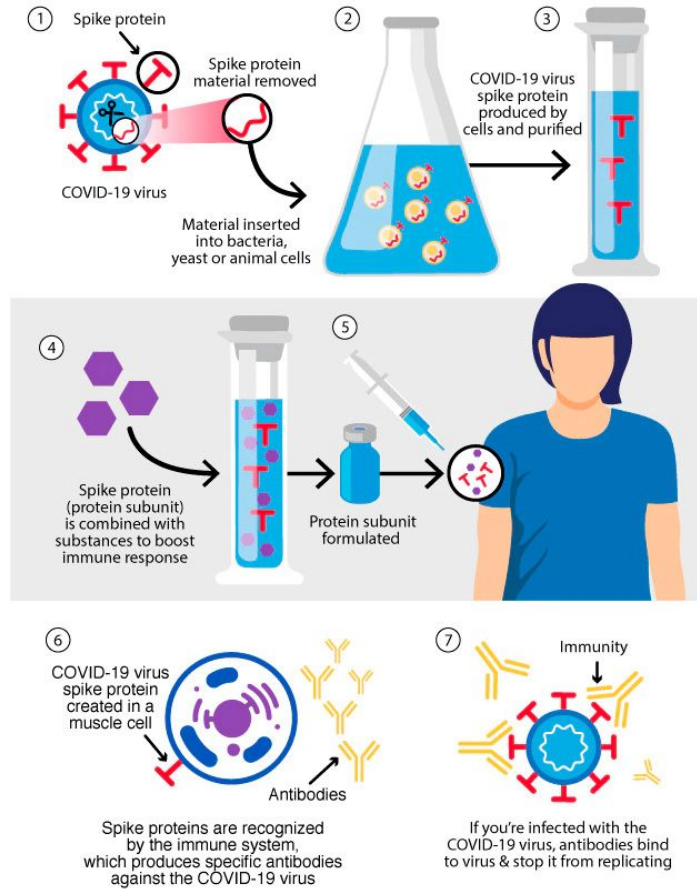


Covid 19 Vaccines



- mRNA
- Viral Vector
- Subunit





Covid 19 Vaccines in India



- Covishield : Oxford and Astra Zeneca : viral vector vaccine
- In this type of vaccine, genetic material from the COVID-19 virus is placed in a modified version of a different virus (viral vector). When the viral vector gets into your cells, it delivers genetic material from the COVID-19 virus that gives your cells instructions to make copies of the S protein. Once your cells display the Spike proteins on their surfaces, your immune system responds by creating antibodies and defensive white blood cells. If you later become infected with the COVID-19 virus, the antibodies will fight the virus.

Covid 19 Vaccines India



- Covaxin : bharat biotech
- Inactivated virus technology
- Sputnik V
- Similar to covishield